Step 1 : Framing Hypotheses

Null Hypothesis $H_0: \mu_1 = \mu_2 = \ldots = \mu_k$

That is, there is no significant difference among the population means of k treatments.

Alternative Hypothesis

$H_1: \mu_i \neq \mu_j$ for atleast one pair (i,j); $i, j = 1,2,\ldots,k$; $i \neq j$

That is, at least one pair of means differ significantly.

Step 2 : Data

Data is presented in the tabular form as described in the previous section

Step 3 : Level of significance : α

Step 4 : Test Statistic

$F = \dfrac{MST}{MSE}$ which follows $F_{(k\text{-}1,\, n\text{-}k)}$, under H_0

To evaluate the test statistic we compute the following:

(i) Correction factor: $C.F = \dfrac{G^2}{n}$ where $G = \sum_{i=1}^{k}\sum_{j=1}^{n_i} x_{ij}$

(ii) Total Sum of Squares: $TSS = \sum_{i=1}^{k}\sum_{j=1}^{n_i} x_{ij}^2 - C.F$

(iii) Sum of Squares between Treatments: $SST = \sum_{i=1}^{k} \dfrac{x_{i.}^2}{n_i} - C.F$,

where $x_{i.} = \sum_{j=1}^{n_i} x_{ij}$, $i = 1,2,\ldots,k$

(iv) Sum of Squares due to Error: $SSE = TSS - SST$

Degrees of Freedom (d.f)

Degrees of freedom (d.f.)		d.f.
Total Sum of Squares	Total no. of observations –1	n–1
Treatment Sum of Squares	Total no. of observations –1	k–1
Error of Sum Squares	Total no. of observations –1	n–k

Mean Sum of Squares

Mean Sum of Squares due to treatment: $MST = \dfrac{SST}{k-1}$

Mean Sum of Squares due to Error:

$$MSE = \frac{SSE}{n-k}$$

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